The Media Connectivity Association




OVERVIEW AND UPDATES · AI IN MEDIA SERVICES

# AI in Media Services

Evaluating AI and ML for media, and the traffic of AI media services

3GPP

NOVEMBER 2026 – www.5g-mag.com

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# About the project

The documents the project rests on

## 3GPP TR 26.927

**Technical Report**

Study on AI and ML in 5G media services: media-based applications and the media service architecture for AI/ML.

## 3GPP TR 26.847

**Technical Report**

Evaluation of AI and ML in 5G media services: feasibility evaluations for a given set of services.

## 3GPP TR 26.870

**Technical Report, draft**

Study on Media Aspects for 6G System, which records the traffic of AI media services measured with the testbed.

FIND OUT MORE **Technology** **Specifications**

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# What the reports cover

The SA4 reports on AI and ML in media, and the 6G media study that the traffic measurements feed

**TR 26.927** media-based applications and the architecture for AI/ML

- Object recognition in image and video
- Video quality enhancement in streaming
- Crowd-sourced media capture
- Natural language processing on speech
- Media service architecture for AI/ML
- Split inference
- Model delivery

**TR 26.847** the evaluation of the services

- Testbed architectures and anchors
- Split inferencing
- Evaluation metrics
- Compressed model transfer, speech recognition
- Object detection and labelling
- Bit-incremental model transmission
- Real-time translation over IMS
- Model-data transmission

**TR 26.870, draft** the seven work topics of the 6G media study; only AI traffic characteristics have findings so far

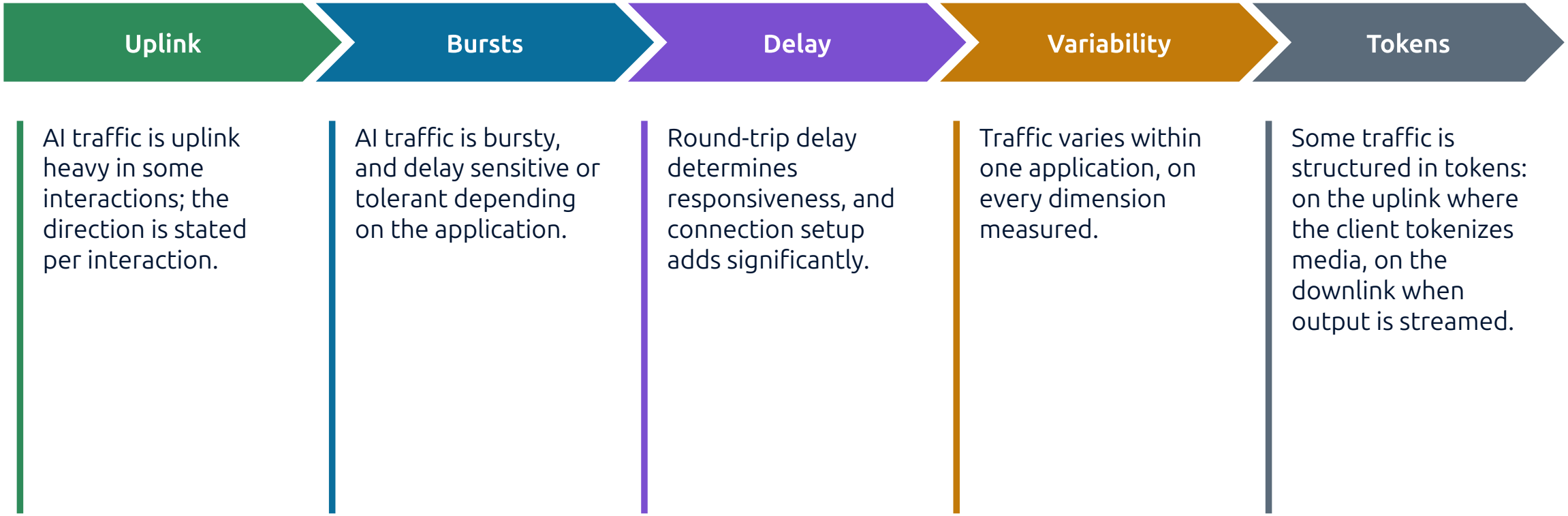
- Media Delivery Architecture
- 6G media
- Media aspects related to SA2 topics
- Media for ubiquitous access
- Trusted and private communication for media
- Software and asset management
- Real-time communication system enhancements

AI/ML in media is studied in SA4; the network-side analytics of AI/ML (NWDAF) are studied in SA2.

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# AI traffic characteristics

The five initial findings of the 6G media study, measured with the 6G AI Traffic Characterization testbed




The findings are initial, from a draft report, and are to be updated as more measurement results become available.

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# The testbed

How the 6G AI Traffic Characterization testbed runs an interaction and measures its traffic

**Orchestrator**

The Orchestrator drives the interactions (chat, agents, image, video, audio) and the MCP tool servers.

**LLM client layer**

Carries the requests to the LLM and tool APIs, with L7 capture by mitmproxy.

**Network emulator**

Applies the tc/netem profiles to the traffic before it reaches the Internet.

**Capture and analysis**

L3/L4 and L7 capture feed the TrafficLogger and the MetricsCalculator.

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# What it measures

Eleven interaction patterns, run over emulated network profiles, reported in four metric categories

## Interaction patterns from chat to agent-to-agent collaboration

- Chat
- Agent with MCP
- Image generation
- Multimodal analysis
- Multimodal analysis, live AI mode
- Video understanding
- Real-time text and audio
- Direct web search
- Computer use
- Personal assistant agent
- Agent-to-agent collaboration

## Network profiles deployment conditions, plus a lossy profile that finds where an interaction breaks

- ideal
- 6g_itu_hrllc
- 5g_urban
- wifi_good
- cell_edge
- satellite_leo
- satellite_geo
- congested
- 5qi_7
- 5qi_80
- lossy

## Metrics recorded per request and from packet captures

- A. Traffic characteristics
- B. Response time and errors
- C. Service metrics
- D. Task-related QoE

A test run takes one interaction pattern, one network profile and one execution mode, and produces a metric record.

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# How it grew

What each 3GPP release added to AI and ML in media services

## Release 18

The SA4 study FS_AI4Media is listed in TR 21.918 without a description. The SA1 work item AIML_MT covers the traffic of AI/ML model transfer.

## Release 19

TR 26.927 and TR 26.847 are approved at SA#108 and created as version 19.0.0.

## Release 20

Work plan of 28 September 2026: the reports recommend stage 3 work on AI/ML in IMS and a new evaluation study. The 6G media study, TR 26.870, is a draft (V0.6.1). Scopes unverified.

From the Standards pages of the website; the Release 20 items come from the 3GPP work plan, with no Release 20 release description (TR 21.920) available.

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# Repositories

The repositories of the AI traffic and AI/ML evaluation work, from their READMEs

| REPOSITORY | WHAT IT IS | COVERS | LATEST RELEASE | |
|---|---|---|---|---|
| 6G-Testbed | the 6G AI Traffic Characterization testbed: measures AI service traffic under network conditions | AI traffic testbed<br>Network emulation | none yet | GitHub |
| netemu-dashboard | a web UI and REST API for network impairment profiles | Network emulator control | none yet | GitHub |
| rt-ai-ml-evaluation-framework | the AI and ML evaluation framework of the 3GPP SA4 report on AI and ML in 5G media services | Compressed model transfer<br>Split inferencing<br>Translation scenario | none yet | GitHub |

FIND OUT MORE Technology Specifications

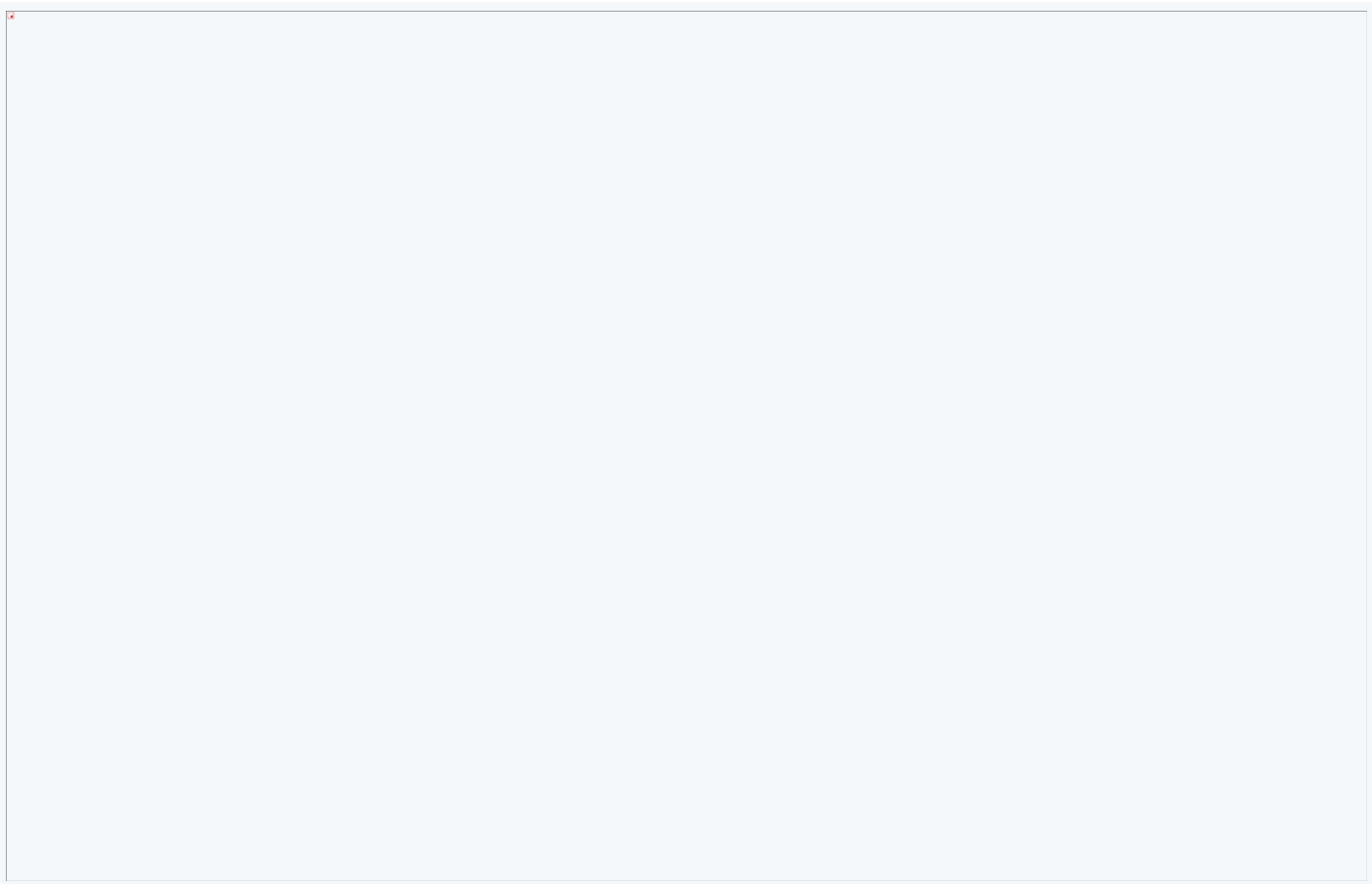
● released ● no release yet

More information at 5g-mag.com/tech/ai-in-media-services

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# Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



PROVIDED BY THE REFERENCE TOOLS

1 **6G-Testbed**

All seven pipeline boxes are in the 6G-Testbed repository (aitestbed).

WHERE WE STAND

# AI in Media Services

## BROUGHT TO STUDY

- More evaluation scenarios: pose and gesture split inferencing, video super resolution, federated learning
- Metrics
- AI gateway
- Agentic uplink

## ON-GOING

- Bit incremental transfer of model data
- Measurement package
- Scenario configurations for video-traffic analysis
- Chat scenario in which the client sends token identifiers

## OUTCOMES

- AI traffic characterization testbed (draft TR 26.870)
- Model transfer and split inference (TR 26.927)
- Real-time translation scenario

RELATION WITH SDOS AND INDUSTRY FORA 3GPP

More information at **5g-mag.com/standards/ai-in-media-services**

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